CENTRE FOR ARTS, DESIGN & INFORMATION TECHNOLOGY

School of Computer Technology

Approved By Associate Dean:

Jan 2, 2025

Signature

COURSE SECTION INFORMATION

Data Visualization Techniques Applied A.I. Solutions Development

Note: All academic inquiries will be replied to within three business days.

COURSE DESCRIPTION:

This course will provide an introduction to data visualisation principles. The focus of this course will be on the tools and techniques of data visualisation. Students will apply current industry best practices while building effective data reports using popular Python and JavaScript libraries and interactive dashboards using tools such as Tableau.

COURSE OUTCOMES:

Upon successful completion of this course the students will have reliably demonstrated the ability to:

1. Explain the data visualisation process and best practices.
2. Communicate insights to stakeholders by building effective data reports and interactive dashboards.
3. Interpret and infer conclusions from data visualisations.
4. Formulate data stories using Python libraries such as Matplotlib, Bokeh, and Seaborn.
5. Create effective data charts using current industry tools such as Tableau.

LIST OF TEXTBOOKS AND OTHER TEACHING AIDS:

Required:
Recommended Resources:
COURSE DELIVERY MODE:
Refer to the table below for the delivery mode.

Important Note on the Use of Generative AI:

Students must review the "Generative AI Usage Guidelines" document, available on D2L, or consult with the instructor for details on how generative AI tools may be used in this course.
Generally, use of AI is allowed and encouraged in most of the courses of this program, this include projects and assignments. However use of AI are not allowed in exams and quizzes unless specified otherwise by the instructor.
Students must consult their instructor when unsure.

Misuse of AI in assessments where it is not permitted or failure to adequately disclose its use will be treated as a violation of academic integrity. According to college policy, consequences may include failing the assignment or the course or more severe disciplinary actions. **Students must also download the AI Usage Declaration form from D2L, complete it, and submit it with their assignments where AI use is permitted.** Adherence to these guidelines is mandatory to maintain academic integrity.

Detailed Evaluation System

Assessment Tool:	Description:	Outcome(s) assessed:	EES assessed:	Date / Week:	Final Grade:
Participation	Attendance and in-class participation in various activities and individual assignments	1, 2, 3, 4, 5	1, 2, 4, 5, 6, 7, 8, 9, 10, 11	1-3	10
Lab Exercises 3 @ 10%	Hands-on exercises	1, 2, 3, 4, 5		1-3	30
Lab quizzes 5 @ 2%	Few questions per class to evaluate students' pace on keeping up with the class			1-3	10
Project	Group Assignment			3	30
Presentation and/or Questionnaire	Project presentation and/or Q&A to evaluate the overall knowledge of the students	1, 2, 3, 4, 5	1,2,4,5,6,7,8,9, 10,11	3	20
				TOTAL	**100%**

GRADING SYSTEM the passing grade for this course is: _D (50%)

A+	90-100	4.0	B+	77-79	3.3	C+	67-69	2.3	D+	57-59	1.3	Below 50	F	0.0
A	86-89	4.0	B	73-76	3.0	C	63-66	2.0	D	50-56	1.0			
A-	80-85	3.7	B-	70-72	2.7	C-	60-62	1.7						

Excerpt from the College Policy on Academic Dishonesty:
The *minimal* consequence for submitting a plagiarized, purchased, contracted, or in any manner inappropriately negotiated or falsified assignment, test, essay, project, or any evaluated material will be a grade of zero on that material.
To view George Brown College policies please go to www.georgebrown.ca/policies

Learning Schedule / Topical Outline (subject to change with notification)

TOPICAL OUTLINE:

Week	Topic Task	In-person / Online	Outcomes	Content / Activities	Resources
1	1	Monday Online	1, 2, 3	- Data and context - Data types (structured and unstructured) - Process Automation, Collaboration, Communication and Research	Resource material available on Brightspace/D2L
1	2	Wednesday Online	1, 2, 3	- Data Warehousing – Dimensional data structures, Star schema - Business Intelligence framework - Data Preparation (ETL/ELT) - Business Intelligence tools: Tableau Data Prep	Resource material available on Brightspace/D2L
1	3	Friday In-person	1, 2, 3	- BI professional tools: Tableau Software - Data visualizations: formatting, tables, parameters, filters, and calculated fields - Chart types (text, Bar H/V, Gantt, Pie, Scatter, geography, histograms, etc.)	Resource material available on Brightspace/D2L
	Week 1 Tasks: Lab Exercise 1 Lab Quiz 1, 2				

2	4	Monday In-person	1, 2, 3, 4	- Key Performance Indicators (KPIs) – Types - Dashboards, Scorecard	Resource material available on

				- Balanced Scorecard	Brightspace/ D2L
2	5	Wednesday Online	1, 2, 3, 4	- Decision making techniques (information visualization, Pareto, Decision Mapping – tree, root cause analysis, statistical analysis, etc.) - Analytics: Trend analysis, Pattern Recognition - Applied decision making using data visualization support systems	Resource material available on Brightspace/ D2L
2	6	Friday In-person	1, 2, 3, 4	- Data visualizations with Python - Creation, rendering, reporting - Use of Python libraries such as Matplotlib, Seaborn, NumPy, etc.	Resource material available on Brightspace/ D2L
	Week 2 Tasks: Lab Exercise 2 Lab Quiz 3, 4				

3	7	Monday In-person	1, 2, 3, 4, 5	- Python's engine Integration with Tableau, use of Python libraries within an ad 'hoc visualization environment to create and render professional python-driven visualizations	Resource material available on Brightspace /D2L
3	8	Wednesday Online	1, 2, 3, 4, 5	- Storytelling techniques - Data Visualization Best Practices - Applied Storytelling – Case Study - Problem Solving using Data Visualization Techniques	Resource material available on Brightspace /D2L
3	9	Friday In-person	1, 2, 3, 4, 5	- Project Due - Presentation and/or Questionnaire	Resource material available on Brightspace /D2L
	Week 3 Tasks: Lab Exercise 3 Lab Quiz 5 Project Due Presentation and/or Questionnaire				

Please note: this schedule may change as resources and circumstances require. For information on withdrawing from this course without academic penalty, please refer to the College Academic Calendar: http://www.georgebrown.ca/Admin/Registr/PSCal.aspx